

This key should allow you to understand why you choose the option you did (beyond just getting a question right or wrong). More instructions on how to use this key can be found [here](#).

If you have a suggestion to make the keys better, please fill out the short survey [here](#).

Note: This key is auto-generated and may contain issues and/or errors. The keys are reviewed after each exam to ensure grading is done accurately. If there are issues (like duplicate options), they are noted in the offline gradebook. The keys are a work-in-progress to give students as many resources to improve as possible.

1. Simplify the expression below into the form $a + bi$. Then, choose the intervals that a and b belong to.

$$\frac{54.00 + 33.00i}{-1.00 + 8.00i}$$

The solution is $3.23 - 7.15i$, which is option E.

- A. $a \in [2.5, 3.5]$ and $b \in [-8, -6]$

$3.23 - 7.15i$, which is the correct option.

- B. $a \in [-54.5, -53.5]$ and $b \in [3, 4.5]$

$-54.00 + 4.12i$, which corresponds to just dividing the first term by the first term and the second by the second.

- C. $a \in [-5.5, -4]$ and $b \in [5.5, 6.5]$

$-4.89 + 6.14i$, which corresponds to forgetting to multiply the conjugate by the numerator and not computing the conjugate correctly.

- D. $a \in [2.5, 3.5]$ and $b \in [-466, -464]$

$3.23 - 465.00i$, which corresponds to forgetting to multiply the conjugate by the numerator.

- E. $a \in [209.5, 210.5]$ and $b \in [-8, -6]$

$210.00 - 7.15i$, which corresponds to forgetting to multiply the conjugate by the numerator and using a plus instead of a minus in the denominator.

General Comment: Multiply the numerator and denominator by the *conjugate* of the denominator, then simplify. For example, if we have $2 + 3i$, the conjugate is $2 - 3i$.

2. Simplify the expression below into the form $a + bi$. Then, choose the intervals that a and b belong to.

$$(5.00 - 2.00i)(-8.00 - 9.00i)$$

The solution is $-58.00 - 29.00i$, which is option E.

- A. $a \in [-59, -56]$ and $b \in [-31, -26]$

$-58.00 - 29.00i$, which is the correct option.

- B. $a \in [-23, -18]$ and $b \in [-63, -59]$

$-22.00 - 61.00i$, which corresponds to adding a minus sign in the first term.

- C. $a \in [-23, -18]$ and $b \in [57, 62]$

$-22.00 + 61.00i$, which corresponds to adding a minus sign in the second term.

- D. $a \in [-59, -56]$ and $b \in [21, 33]$

$-58.00 + 29.00i$, which corresponds to adding a minus sign to both term.

E. $a \in [-45, -37]$ and $b \in [15, 19]$

$-40.00 + 18.00i$, which corresponds to just multiplying the real terms to get the real part of the solution and the coefficients in the complex terms to get the complex part.

General Comment: You can treat i as a variable and distribute. Just remember that $i^2 = -1$, so you can continue to reduce after you distribute.

3. Simplify the expression below and choose the interval the simplification is contained within.

$$8 - 11 \div 20 * 14 - (7 * 17)$$

The solution is -118.7 , which is option A.

A. $[-120.1, -118.3]$

-118.7 , which is the correct option.

B. $[-111.9, -109.5]$

-111.039 , which corresponds to an Order of Operations error: not reading left-to-right for multiplication/division.

C. $[125, 127.2]$

126.961 , which corresponds to not distributing addition and subtraction correctly.

D. $[-114.8, -111.8]$

-113.9 , which corresponds to not distributing a negative correctly.

E. None of the above

You may have gotten this by making an unanticipated error. If you got a value that is not any of the others, please let the coordinator know so they can help you figure out what happened.

General Comment: While you may remember (or were taught) PEMDAS is done in order, it is actually done as P/E/MD/AS. When we are at MD or AS, we read left to right.

4. Choose the **smallest** set of Real numbers that the number below belongs to.

$$\sqrt{\frac{53361}{441}}$$

The solution is Whole, which is option C.

A. Whole

* This is the correct option!

B. Integer

These are the negative and positive counting numbers (... , -3, -2, -1, 0, 1, 2, 3, ...)

C. Rational

These are numbers that can be written as fraction of Integers (e.g., $-2/3$)

D. Irrational

These cannot be written as a fraction of Integers.

E. Not a Real Number

These are Nonreal Complex numbers **OR** things that are not numbers (e.g., dividing by 0).

General Comment: First, you **NEED** to simplify the expression. This question simplifies to 231.

Be sure you look at the simplified fraction and not just the decimal expansion. Numbers such as 13, 17, and 19 provide **long but repeating/terminating decimal expansions!**

The only ways to *not* be a Real number are: dividing by 0 or taking the square root of a negative number.

Irrational numbers are more than just square root of 3: adding or subtracting values from square root of 3 is also irrational.

5. Choose the **smallest** set of Complex numbers that the number below belongs to.

$$\sqrt{\frac{-1989}{0}}i + \sqrt{234}i$$

The solution is Not a Complex Number, which is option C.

- A. Rational

These are numbers that can be written as fraction of Integers (e.g., $-2/3 + 5$)

- B. Irrational

These cannot be written as a fraction of Integers. Remember: π is not an Integer!

- C. Nonreal Complex

This is a Complex number ($a + bi$) that is not Real (has i as part of the number).

- D. Pure Imaginary

This is a Complex number ($a + bi$) that **only** has an imaginary part like $2i$.

- E. Not a Complex Number

* This is the correct option!

General Comment: Be sure to simplify $i^2 = -1$. This may remove the imaginary portion for your number. If you are having trouble, you may want to look at the *Subgroups of the Real Numbers* section.
